

Daily Review

Lesson 2

- 1 What does a social impact affect?
- 2 “More tourists spend money” – type?
- 3 A negative effect of building work?
- 4 A real Scottish renewable project?
- 5 A strong impact answer says...?

People's lives

Economic

Habitats damaged, or
waste

Whitelee or Pitlochry
Dam

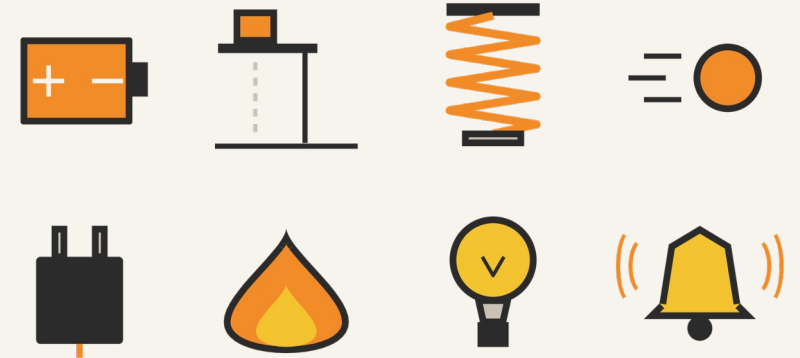
What, why and who

TEST 1 REVISION • LESSON 3 OF 4

Energy In, Energy Out

Where does the energy go?

S3 Engineering Science



THE BIG QUESTION

Whitelee's turbines don't make energy. Where does it come from — and go?



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PART 1

Energy In, Energy Out

Name the forms, then follow the energy through a system.

By the end of this part...

- 1 I can name forms of energy
- 2 I can describe energy transfers: input, useful, wasted
- 3 I can identify useful and wasted energy
- 4 I can draw a simple energy transfer diagram
- 5 I can explain my ideas clearly using what + why

SECTION A

Eight Forms of Energy

A battery, a skier, a spring — what do they have in common?

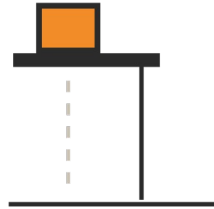
Three Energy Stores

Energy waiting to be used



Chemical

in fuel, food and
batteries



**Gravitational
potential**

anything high up



Elastic

springs, a drawn bow

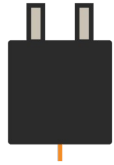
Five Forms on the Move

Energy is measured in joules (J)



Kinetic

moving
things



Electrical

circuits,
the mains



Heat

makes
things
hotter



Light

lamps, the
Sun



Sound

vibrations

Worked Example – A Skier

One object, several forms of energy

1 Top

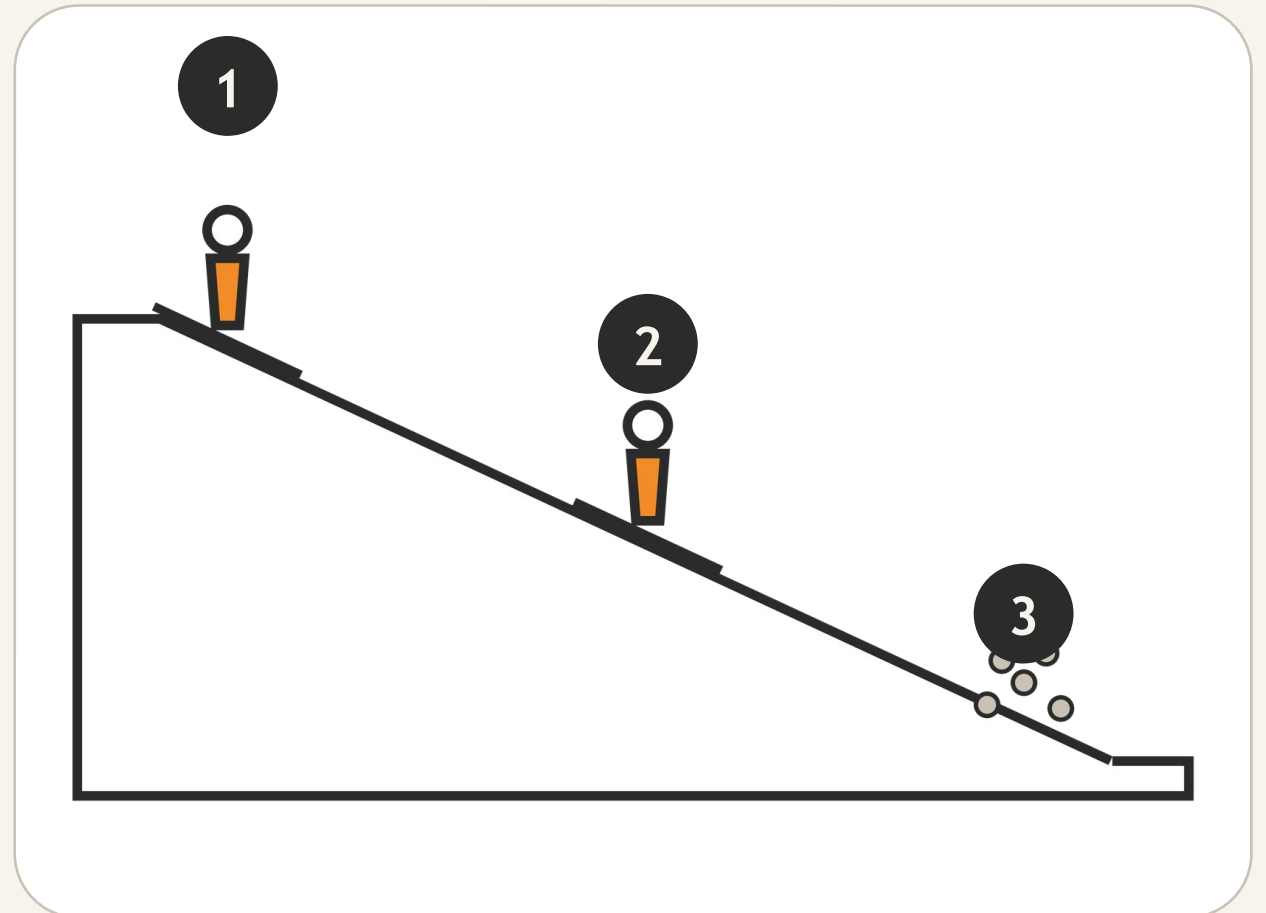
gravitational potential

2 Skiing down

kinetic

3 The snow

heat and sound – wasted



Show Me — Which Form?

SHOW ME BOARDS

- 1 A full petrol tank
- 2 A book on a high shelf
- 3 A squashed spring in a clothes peg
- 4 A spinning washing-machine drum
- 5 The unit of energy

Chemical

Gravitational potential

Elastic

Kinetic

Joules (J)

Show Me — Which Form?

SHOW ME BOARDS

EXTRA PRACTICE

- 1 A radiator warming a room
- 2 A school bell ringing
- 3 A drawn bow before the arrow flies
- 4 Two forms for a moving train
- 5 Two forms for a charging phone

Heat

Sound

Elastic

Kinetic, sound (and
heat)

Electrical, chemical

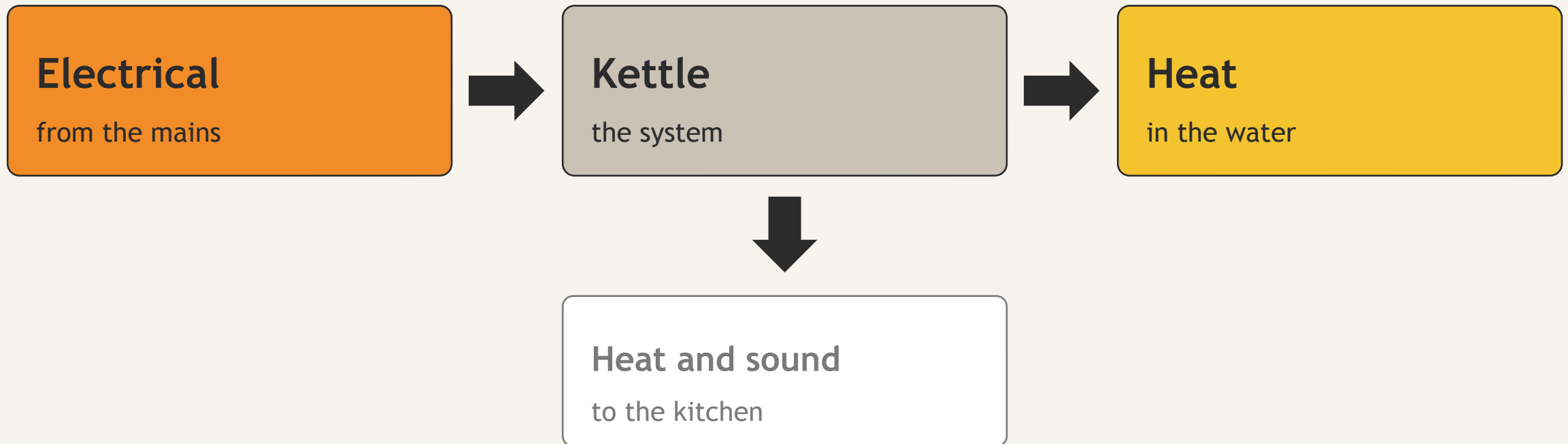
SECTION B

Where Does It Go?

Energy is never used up. So where does it go?

Input, Useful, Wasted – Electric Kettle

Energy is transferred, never used up



Worked Example – The Input Is What Goes IN

Trap: electrical is the OUTPUT of a turbine, not the input

Input

kinetic energy of the wind

Useful output

electrical

Wasted

sound and heat



Useful Depends on the Job

The same form can be useful in one device and wasted in another

Heat

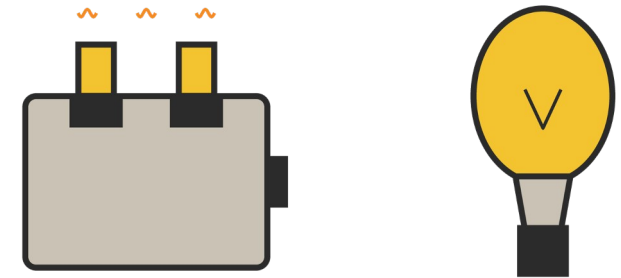
useful in a toaster · wasted in a lamp

Light

useful in a lamp · wasted in a toaster

Sound

useful in a smoke alarm · wasted in a kettle



Show Me — Input, Useful, Wasted

SHOW ME BOARDS

1 Kettle: the input energy?

Electrical

2 Lamp: the useful output?

Light

3 Lamp: one wasted form?

Heat

4 Wind turbine: the input energy?

Kinetic (of the wind)

5 What happens to wasted energy?

Spreads out as heat

Show Me – Input, Useful, Wasted

SHOW ME BOARDS

EXTRA PRACTICE

- 1 Petrol car: the input energy?
- 2 Petrol car: the useful output?
- 3 Toaster: is the light useful or wasted?
- 4 Smoke alarm: is the sound useful or wasted?
- 5 Why does wasted energy still cost money?

Chemical (in the petrol)

Kinetic

Wasted

Useful

You paid for the input

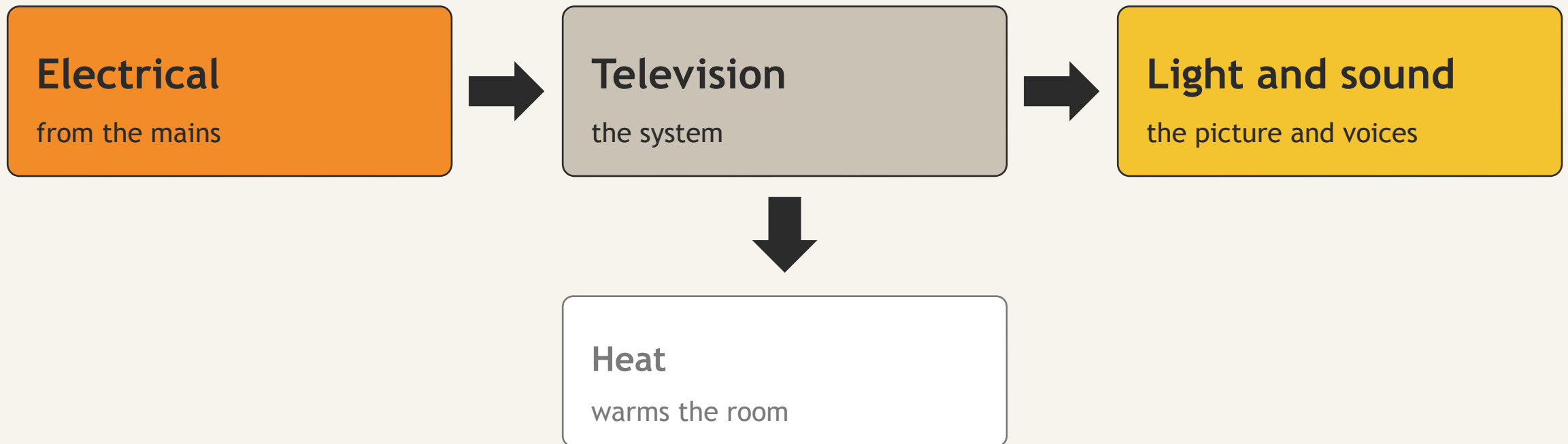
SECTION C

Draw It

Two marks for a diagram. Where does the second mark go?

Worked Example – A Television

Mark 1: input + useful. Mark 2: wasted on its own arrow.



Where the Marks Go – A Laptop

A – 2 marks

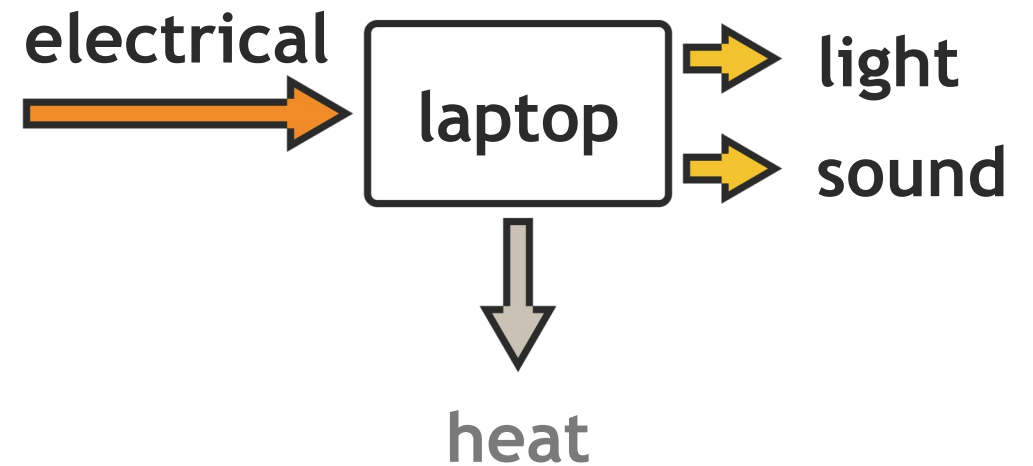
heat on its own wasted arrow

B – 1 mark

heat drawn as a useful output

C – 0 marks

only “electrical → laptop”



Show Me — Draw It

SHOW ME BOARDS

Draw the diagram on your board.

- 1 Toaster
- 2 Phone charger
- 3 Petrol car
- 4 Candle
- 5 How do you show the wasted energy?

Electrical → heat; light
wasted

Electrical → chemical;
heat wasted

Chemical → kinetic;
heat, sound wasted

Chemical → light; heat
wasted

Its own arrow, going
down

Show Me — Draw It

SHOW ME BOARDS

EXTRA PRACTICE

Draw the diagram on your board.

- 1 Electric kettle
- 2 Television (from memory)
- 3 Wind turbine
- 4 Gas cooker
- 5 Why a separate wasted arrow?

Electrical → heat; sound
wasted

Electrical → light,
sound; heat wasted

Kinetic → electrical;
sound wasted

Chemical → heat; light
wasted

It isn't the job — mark 2

Finger Vote – Common Mix-ups

FINGER VOTE

Think first. Fingers on your chest – show on 3, 2, 1.

What energy does a battery store?

- 1 electrical
- 2 chemical
- 3 kinetic
- 4 light

Answer 2

It stores chemical energy. Electrical is what comes out.

Finger Vote – Common Mix-ups

FINGER VOTE

Think first. Fingers on your chest – show on 3, 2, 1.

Where does wasted energy go?

- 1 it is used up
- 2 it disappears
- 3 it spreads into the surroundings as heat
- 4 it turns back into electrical

Answer 3

Energy is never used up – it spreads out, mostly as heat.

Finger Vote – Common Mix-ups

FINGER VOTE

Think first. Fingers on your chest – show on 3, 2, 1.

What is the INPUT energy of a wind turbine?

- 1 electrical
- 2 gravitational potential
- 3 sound
- 4 kinetic

Answer 4

The wind's kinetic energy goes in. Electrical comes out.

Finger Vote – Common Mix-ups

FINGER VOTE

Think first. Fingers on your chest – show on 3, 2, 1.

The sound from a smoke alarm is...

- 1 useful
- 2 wasted
- 3 both
- 4 not energy

Answer 1

Warning people is its job, so the sound is useful here.

Finger Vote – Common Mix-ups

FINGER VOTE

Think first. Fingers on your chest – show on 3, 2, 1.

Which lamp diagram gets both marks?

- 1 electrical → light + heat, both useful
- 2 electrical → light; heat on its own arrow
- 3 electrical → lamp
- 4 light → lamp → electrical

Answer 2

Only 2 shows the heat as wasted, on its own arrow.

Mini Test 3

On your own, in silence, pen only.
10 minutes – 8 marks.
Then swap and mark from the next slides.

Mini Test 3 – Energy In, Energy Out

Mini Test 3 – Marking

ANSWERS

Swap sheets. Tick or cross, then total out of 8.

Q1 (1 + 1)

(a) chemical – not electrical. (b) gravitational potential – not just “potential”.

Q2 (1 + 1)

(a) chemical (in the petrol). (b) kinetic – not “movement”.

Mini Test 3 — Marking

ANSWERS

A bare “light” in Q3(b) scores 0 — say why.

Q3 (1 + 1)

(a) heat. (b) light or sound, because it doesn't help to brown the bread.

Q4 (1 + 1)

Mark 1: electrical in, sound useful out. Mark 2: heat on its own wasted arrow.

Heat drawn as useful: mark 2 = 0.

Plenary

SHOW ME BOARDS

- 1 Name the three energy stores.
- 2 The input energy of a petrol car?
- 3 Heat: useful in one device, wasted in another?
- 4 Draw a toaster's diagram.
- 5 Is energy ever used up?

Chemical, grav.
potential, elastic

Chemical

Toaster / lamp

Electrical → heat; light
wasted

No — it is transferred

Plenary

SHOW ME BOARDS

EXTRA PRACTICE

1 The unit of energy?

Joules (J)

2 Form stored in a stretched spring?

Elastic

3 The two most common wasted forms?

Heat and sound

4 What does the second diagram mark need?

Wasted on its own arrow

5 Which I Can are you least sure of?

Your own answer

LOOKING AHEAD

Lesson 4 — Efficiency and Power

Efficiency, power, the marble run and teamwork.